Scope: International Departures @ Terminal 3 on June 21, 2026

flight	n	route	dest	cap	dep
AI 173	1	DEL-CCU-SFO	SFO	288	0005
AI 312	2	DEL-ICN	ICN	259	0005
VJ 972	3	DEL-HAN	HAN	377	0005
AI 187	4	DEL-VIE-YYZ	YYZ	328	0010
AI 161	5	DEL-LHR	LHR	316	0030
AI 2336	6	DEL-BKK	BKK	164	0030
AI 2118	7	DEL-SIN	SIN	188	0040
AF 283	8	DEL-CDG	CDG	328	0050
AI 302	9	DEL-SYD	SYD	259	0055
AI 277	10	DEL-CMB	CMB	182	0100

Table 1: Excerpt from Corpus, from 0000 hours to 0100 hours

The Corpus contains 164 flights with total capacity 39,610.

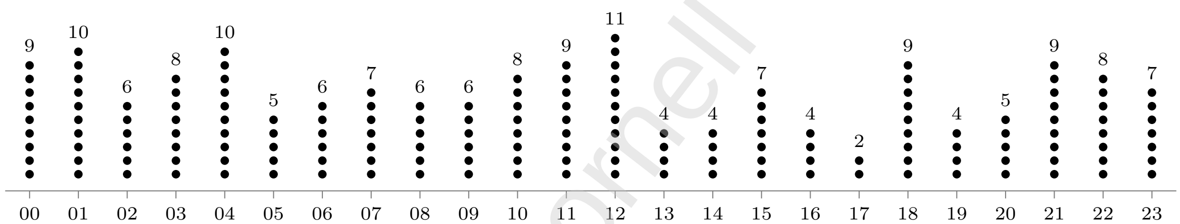

Figure 1: Distribution of dep(flights) over hourly time, each dot = one flight

We assign a serial number to each occupiable seat, continuous over all flights in order of departure.

seat	n	flight
1	1	AI 173
289	2	AI 312
548	3	VJ 972
925	4	AI 187
1253	5	AI 161
1569	6	AI 2336
1733	7	AI 2118
1921	8	AF 283
2249	9	AI 302
2508	10	AI 277

Table 2: Excerpt from Seats, first seat for flights n = [1, 10]

The actual number of Passengers on a flight is defined by its Load Factor, where

Equation 1. Number of passengers: `pax = round(LF * cap)` [for a given flight n].

Assumption 1. *Homogeneous LF: all flights have an equal load factor that is not idiosyncratic; this will be relaxed later.*

For a given seat, we assign a binary quality `exists = {0, 1}`, where a pax is defined by `exists(seat) = 1`.

A flight's check-in window opens at `(dep - 0400)` and closes at `(dep - 0100)`. Each correspondent pax has a time `entry(pax)`, defined as the time a passenger enters the check-in area.

Check-in desks are 'points' that process pax at a given rate.

Assumption 2. *Homogeneous Check-in Rate: all check-in desks have an equal rate of processing that is not idiosyncratic; this will be relaxed later.*

Processing rates are expressed as a service time τ in seconds per pax, or equivalently a service rate $\mu = 1/\tau$ in pax per unit time.

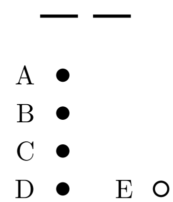

Illustration 1. Basic Setup with 1 desk and 4+1 pax

In the shown arrangement, with a service time of τ seconds per pax, passenger E would queue for 4τ seconds and be processed within 5τ seconds[1].

Assumption 3. *Given multiple check-in options, the rational passenger chooses that with the shortest queue and thence the shortest processing time.*

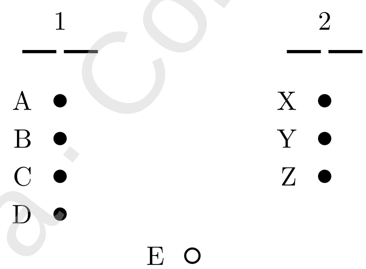

Illustration 2. Rational Choice: Passenger E chooses to queue for check-in desk 2, ceteris paribus.

[1] The two bars represent one check-in desk point, where filled dots represent queued passengers and the empty dot represents a passenger yet to queue.